

FOOTHILLS, NC, USA

WMO: 723148

Lat: 35.821N

Lon: 81.611W

Elev: 387

StdP: 96.76

Time Zone: -5.00 (NAE)

Period: 01-19

WBAN: 63859

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-6.1	-20.1	0.7	0.8	-17.3	0.9	0.9	8.9	8.1	8.1	5.9	1.0	300	0.369

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.1	33.9	21.7	32.6	21.7	31.4	21.6	24.7	29.0	24.1	28.6	23.5	27.9	2.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.5	26.0	22.9	18.4	25.4	22.5	18.0	25.0	76.9	29.4	74.7	28.0	72.2	27.9	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	5.7	4.8	-13.8	35.5	3.6	1.1	-16.4	36.3	-18.5	37.0	-20.5	37.6	-23.1	38.4		
			-14.3	25.7	3.3	0.8	-16.6	26.3	-18.6	26.8	-20.4	27.3	-22.8	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	4.0	5.5	9.8	14.9	19.1	23.3	24.9	24.3	20.8	14.6	9.2	5.8
	DBStd	8.42	5.57	4.71	5.04	4.30	3.63	2.40	2.14	2.31	3.12	4.32	4.33	4.51
	HDD10.0	636	199	141	67	9	0	0	0	0	0	10	64	146
	HDD18.3	2033	445	361	265	118	37	1	0	1	14	129	273	389
	CDD10.0	2362	13	13	63	156	282	398	462	442	323	153	42	15
	CDD18.3	716	0	1	2	14	61	149	204	184	86	14	1	0
	CDH23.3	6659	4	3	42	263	645	1413	1898	1556	687	132	16	1
	CDH26.7	2550	1	0	3	61	197	556	847	644	220	21	1	0
Wind	WSAvg	1.3	1.7	1.7	1.9	1.9	1.4	1.1	0.9	0.9	1.0	1.2	1.2	1.2
Precipitation	PrecAvg	1258	104	105	117	92	107	120	108	104	111	99	93	94
	PrecMax	1670	225	213	318	212	278	276	238	239	302	326	267	212
	PrecMin	814	8	6	29	6	20	16	13	17	2	0	26	2
	PrecStd	194	46	53	56	50	52	53	57	57	74	73	48	50
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.2	23.0	27.2	30.9	32.7	34.7	36.0	35.2	32.9	28.8	26.1	22.2
		MCWB	13.8	15.8	15.0	15.9	19.2	21.2	21.6	22.5	20.1	19.6	16.2	14.6
	2%	DB	18.0	19.8	24.1	28.3	30.8	32.9	33.9	33.7	31.2	27.0	22.4	18.6
		MCWB	12.5	13.5	14.0	15.8	19.0	21.7	22.2	22.1	20.6	17.7	14.5	12.9
	5%	DB	16.0	17.3	22.2	26.9	28.8	31.6	32.6	32.1	29.1	24.7	20.0	16.2
		MCWB	11.3	10.2	13.5	15.6	18.9	21.4	22.1	22.5	20.8	17.1	13.5	12.0
	10%	DB	12.9	14.1	19.0	24.1	27.3	30.0	31.3	30.2	27.4	22.5	17.7	13.4
		MCWB	8.4	8.2	11.9	14.7	18.8	21.0	22.2	22.0	20.2	16.4	12.6	9.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.7	17.6	17.4	19.9	23.0	24.5	25.5	25.8	24.3	22.1	19.2	18.0
		MCDB	19.2	21.6	22.8	24.5	27.3	28.9	29.6	29.9	26.1	25.2	21.0	19.8
	2%	WB	14.6	15.0	16.1	18.6	21.8	23.8	24.6	25.0	23.4	20.7	17.2	15.2
		MCDB	17.2	18.1	20.5	23.9	26.0	28.4	29.5	29.2	26.2	23.3	19.3	16.8
	5%	WB	12.1	11.9	15.1	17.5	21.0	23.1	24.0	24.2	22.5	19.4	15.6	12.7
		MCDB	14.8	15.2	19.8	22.9	25.5	28.1	29.1	28.5	25.9	22.5	18.8	14.6
	10%	WB	9.7	9.5	13.6	16.5	20.2	22.5	23.5	23.5	21.6	17.8	13.8	10.3
		MCDB	12.4	13.1	17.8	21.5	24.6	27.2	28.2	27.6	25.0	20.5	16.9	12.4
Mean Daily Temperature Range	5% DB	MDBR	12.0	12.2	13.3	14.4	13.3	12.7	12.1	11.7	11.5	13.1	14.0	11.5
		MCDBR	15.2	16.9	17.1	18.2	16.4	15.4	14.6	14.4	14.6	15.8	17.3	15.4
		MCWBR	9.5	10.2	8.4	7.4	6.7	5.4	5.0	5.0	5.7	7.9	9.7	9.3
	5% WB	MCDBR	12.5	13.4	13.6	13.4	12.7	12.3	11.9	11.1	10.7	11.2	11.7	11.5
		MCWBR	9.2	9.9	8.0	6.8	6.2	5.3	4.7	4.6	5.1	6.8	8.4	8.5
Clear-Sky Solar Irradiance	taub	0.299	0.307	0.338	0.388	0.436	0.466	0.509	0.502	0.399	0.355	0.314	0.299	
	taud	2.532	2.502	2.455	2.314	2.200	2.151	2.059	2.057	2.333	2.449	2.529	2.562	
	Ebn at Noon	907	939	933	895	853	824	785	782	856	870	881	886	
	Edh at Noon	79	92	105	127	145	152	166	163	118	96	79	73	
All-Sky Solar Radiation	RadAvg	2.52	3.17	4.24	5.27	5.83	6.21	5.78	5.37	4.58	3.74	2.83	2.17	
	RadStd	0.16	0.36	0.28	0.41	0.46	0.44	0.41	0.26	0.42	0.45	0.25	0.20	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (52 neighbors)	+0.51	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+154	+98

Nomenclature: See separate page